Exploring CO2 Emissions

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Presentation Overview

This is an exploratory data analysis of eight countries (Australia, Brazil, Canada, China, India, South Africa, United States of America, and Germany) regarding CO2 emissions spanning from the years 2000-2022.

We will be looking at the amount of CO2 emissions emitted by the people living in these countries as well as CO2 emissions emitted by specific sectors and industries.

Furthermore, we will explore possible correlations between CO2 emissions and sectoral GDP, urbanization, industrial growth, transportation growth, energy consumption, and renewable energy usage.

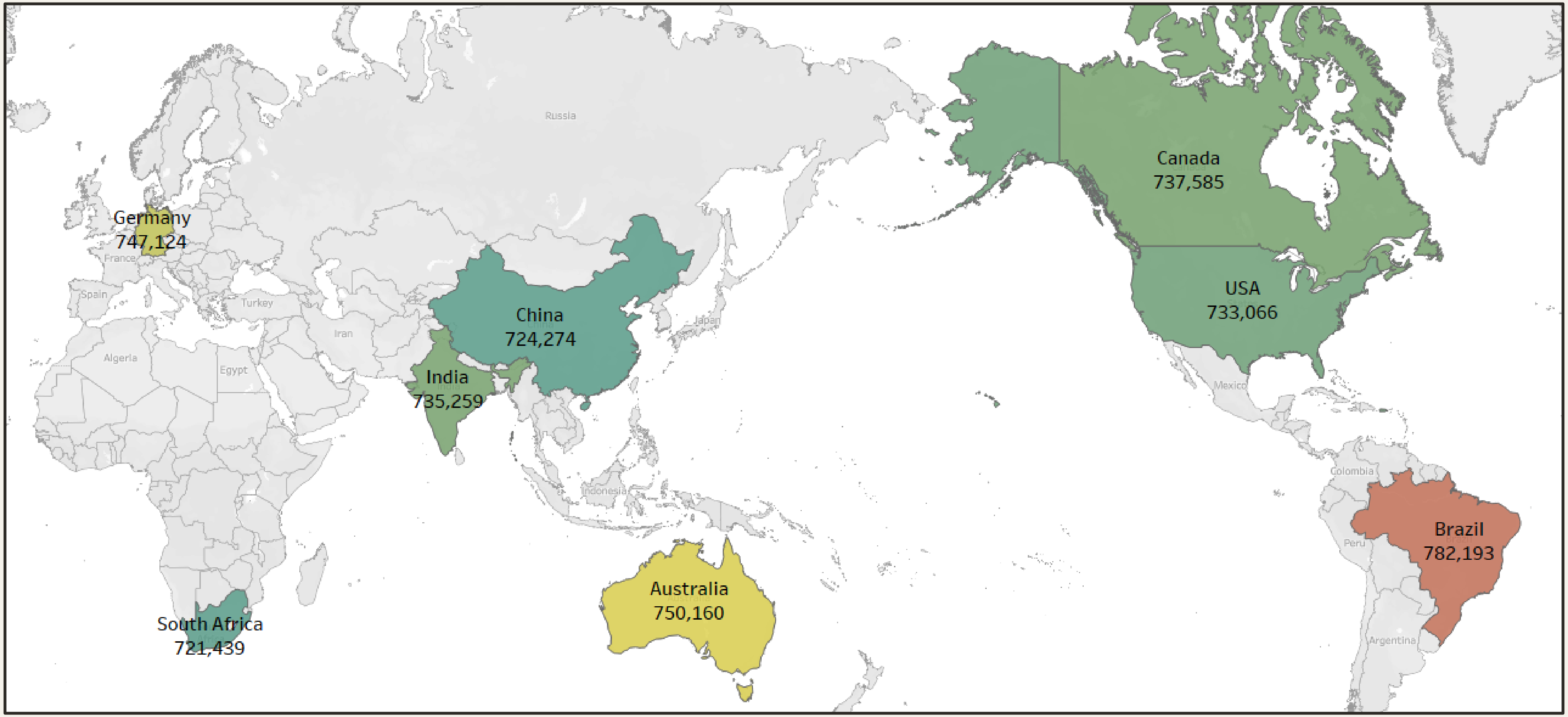
PRESENTATION CONTENT

- CO2 Emissions by Country
- CO2 Emissions by Year
- CO2 Emissions Per Person
- CO2 Emissions by Sector
- CO2 Emissions by Industry
- CO2 Correlations

CO2 EMISSIONS BY COUNTRY

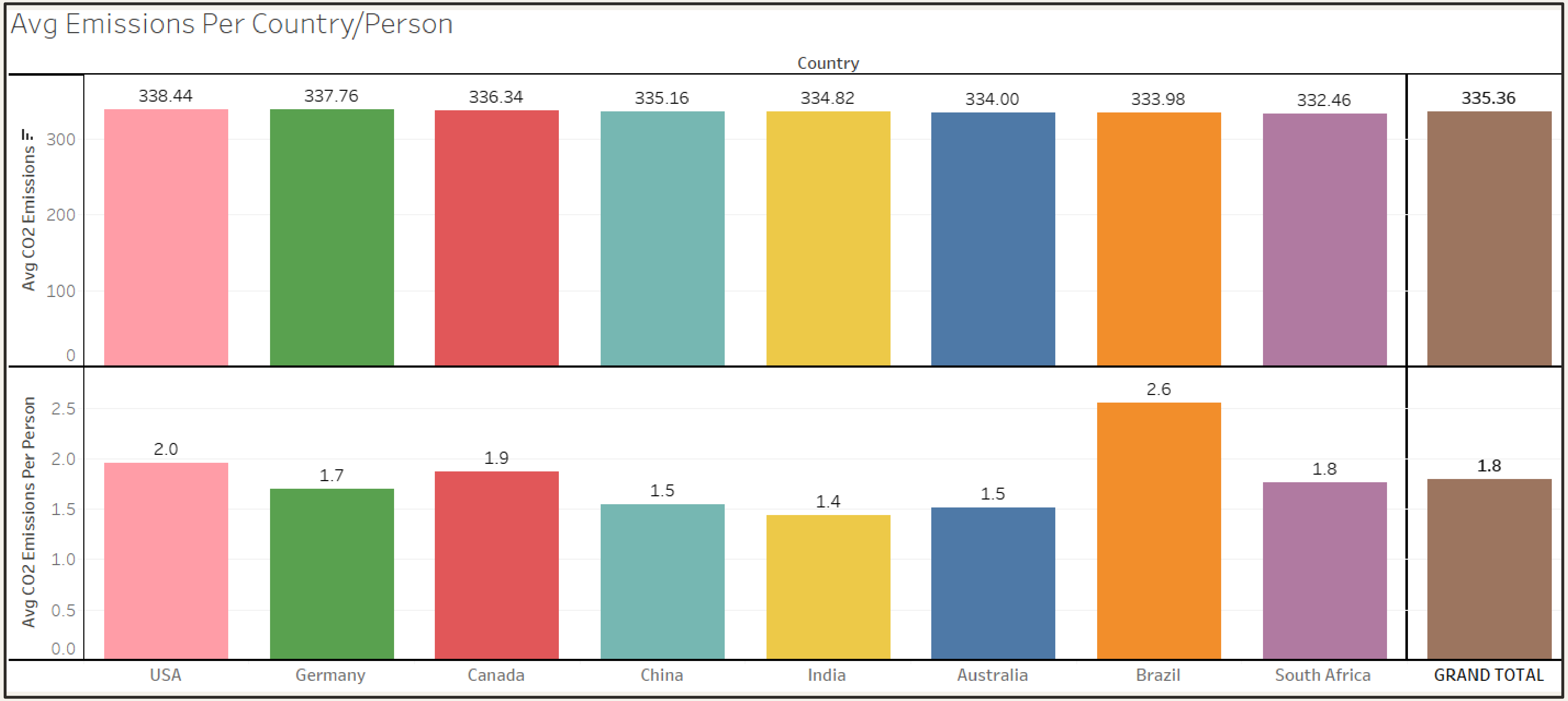
OVERALL CO2 EMISSIONS BY COUNTRY

The country with the most CO2 emissions was Brazil at 782,193 metric tons the country with the least CO2 emissions was South Africa at 721,439 metric tons.



CO2 EMISSIONS BY COUNTRY ON AVERAGE

The country with the most CO2 emissions on average is the USA at 338.44 metric tons per instance and the country with the least CO2 emissions on average is South Africa at 332.46 metric tons per instance.



CO_2 EMISSIONS BY YEAR

CO2 EMISSIONS BY YEAR

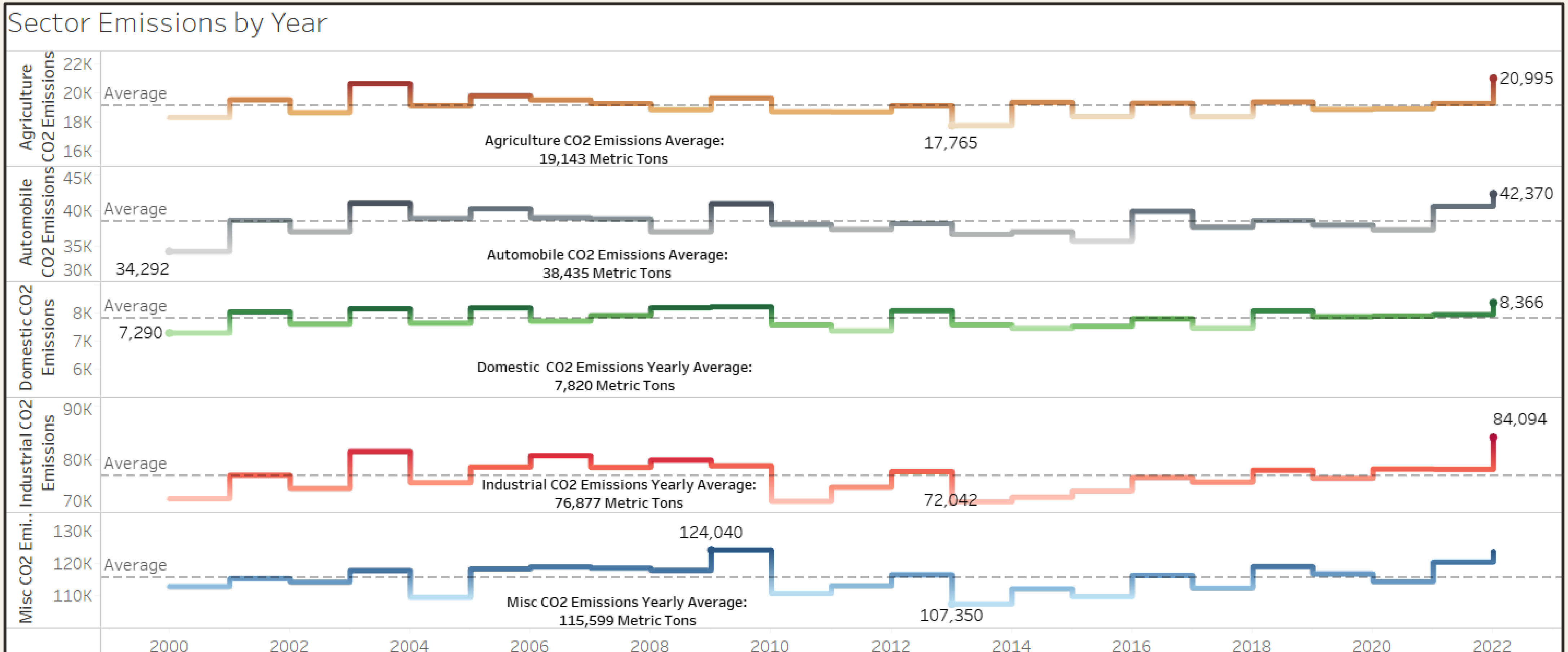
The year with the most CO_2 Emissions was 2022 at 279,407 metric tons and the year with the Least CO_2 Emissions was 2013 at 241,439 metric tons.

Total Emissions by Year

Country	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Australia	30,092	33,418	29,015	33,625	33,354	35,214	36,328	33,001	30,498	32,447	33,142	32,402	34,721	30,352	33,726	34,153	30,32
Brazil	31,987	39,998	34,286	40,926	31,851	32,956	34,113	31,337	36,870	36,966	32,369	32,113	33,688	33,199	37,292	32,063	31,72
Canada	35,738	28,434	32,456	31,837	32,067	31,317	31,815	29,636	32,759	37,673	32,366	29,065	26,578	31,758	34,800	33,573	31,31
China	25,654	29,826	33,121	31,092	28,697	35,051	28,374	40,334	32,760	38,962	26,915	29,926	31,494	29,525	26,324	26,383	32,52
Germany	32,886	38,551	33,318	32,305	29,237	30,311	34,334	30,807	33,050	29,635	28,080	30,964	34,219	29,025	27,115	32,868	36,51
India	30,734	30,342	27,813	26,587	39,941	34,643	33,713	32,278	35,605	32,513	31,784	34,065	34,538	27,019	28,600	28,315	37,97
South Africa	29,463	25,916	33,352	35,023	29,976	30,741	36,592	27,900	25,915	29,513	27,889	37,268	34,084	28,357	36,476	32,080	28,83
USA	28,774	32,084	28,667	37,680	25,589	34,838	30,536	37,715	34,268	33,941	34,533	25,385	30,242	32,203	24,516	25,925	30,65
GRAND TOTAL	245,327	258,569	252,028	269,075	250,712	265,071	265,805	263,010	261,725	271,649	247,079	251,188	259,564	241,439	248,849	245,361	259,87

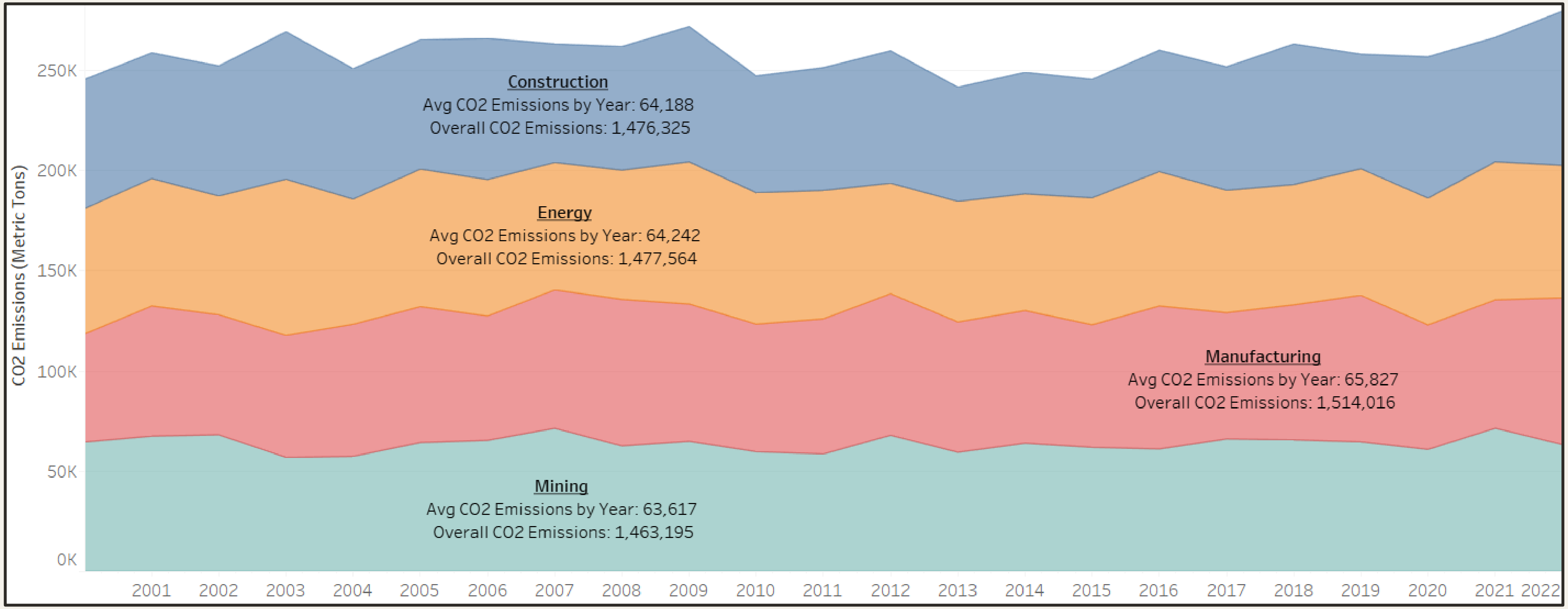
SECTOR CO2 EMISSIONS BY YEAR

Miscellaneous CO2 emissions (commercial activities and public services) emitted the most emissions by sector overall (2,658,766 metric tons) and on average (115,599 metric tons a year). Domestic CO2 emissions emitted the least CO2 emissions by sector overall (179,867 metric tons) and on average (7,820 metric tons a year).



INDUSTRY CO2 EMISSIONS BY YEAR

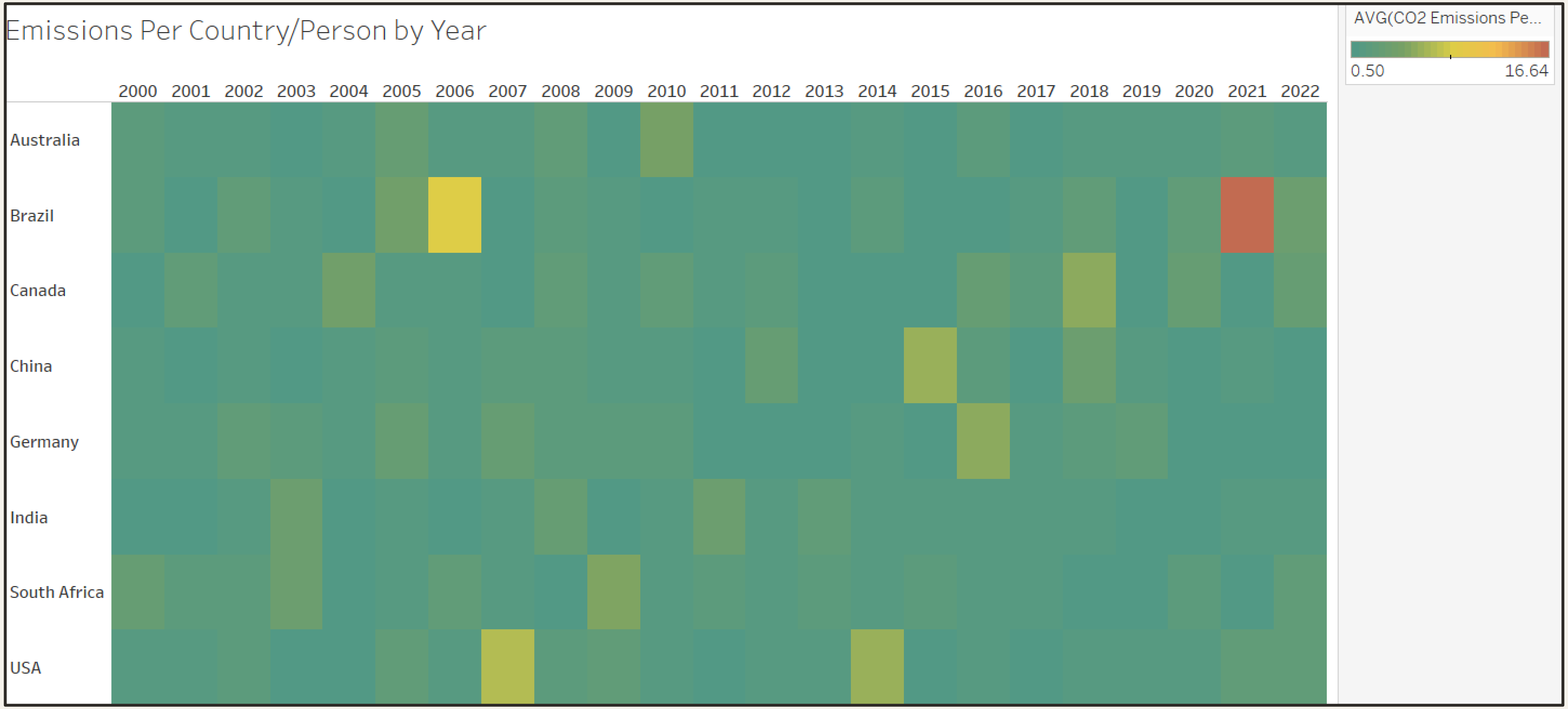
The manufacturing industry creates the most CO2 emissions overall (1,514,016 metric tons) and on average per year (65,827 metric tons). The mining industry creates the least amount of CO2 emissions overall (1,463,195 metric tons) and on average per year (63,617 metric tons).



CO2 EMISSIONS PER PERSON

CO2 EMISSIONS PER PERSON

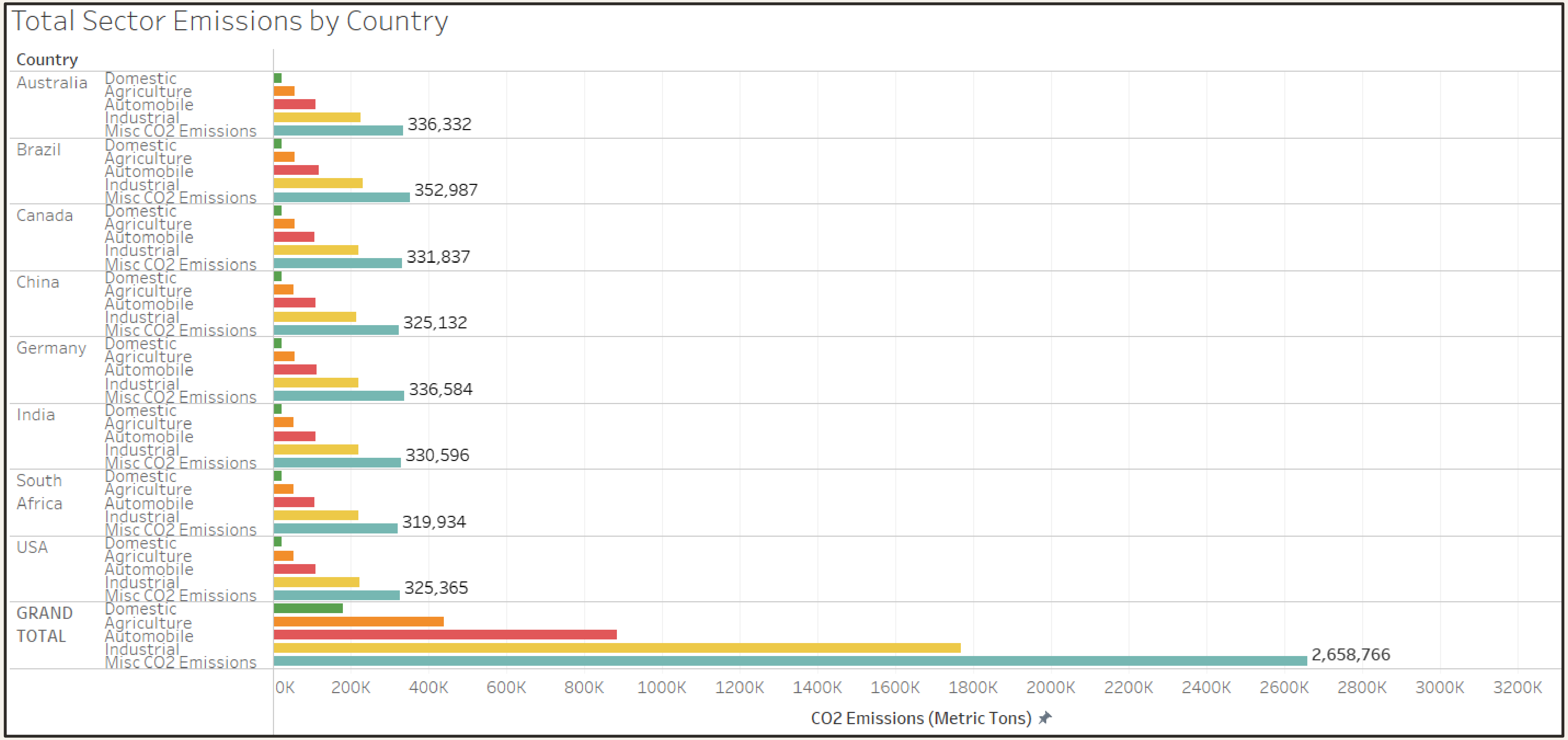
The average person emitted 1.8 metric tons of CO2 emissions. Brazil averaged the most CO2 emissions per person at 2.6 metric tons while India produced the least amount of CO2 emissions per person at 1.4 metric tons.



CO2 EMISSIONS BY SECTOR

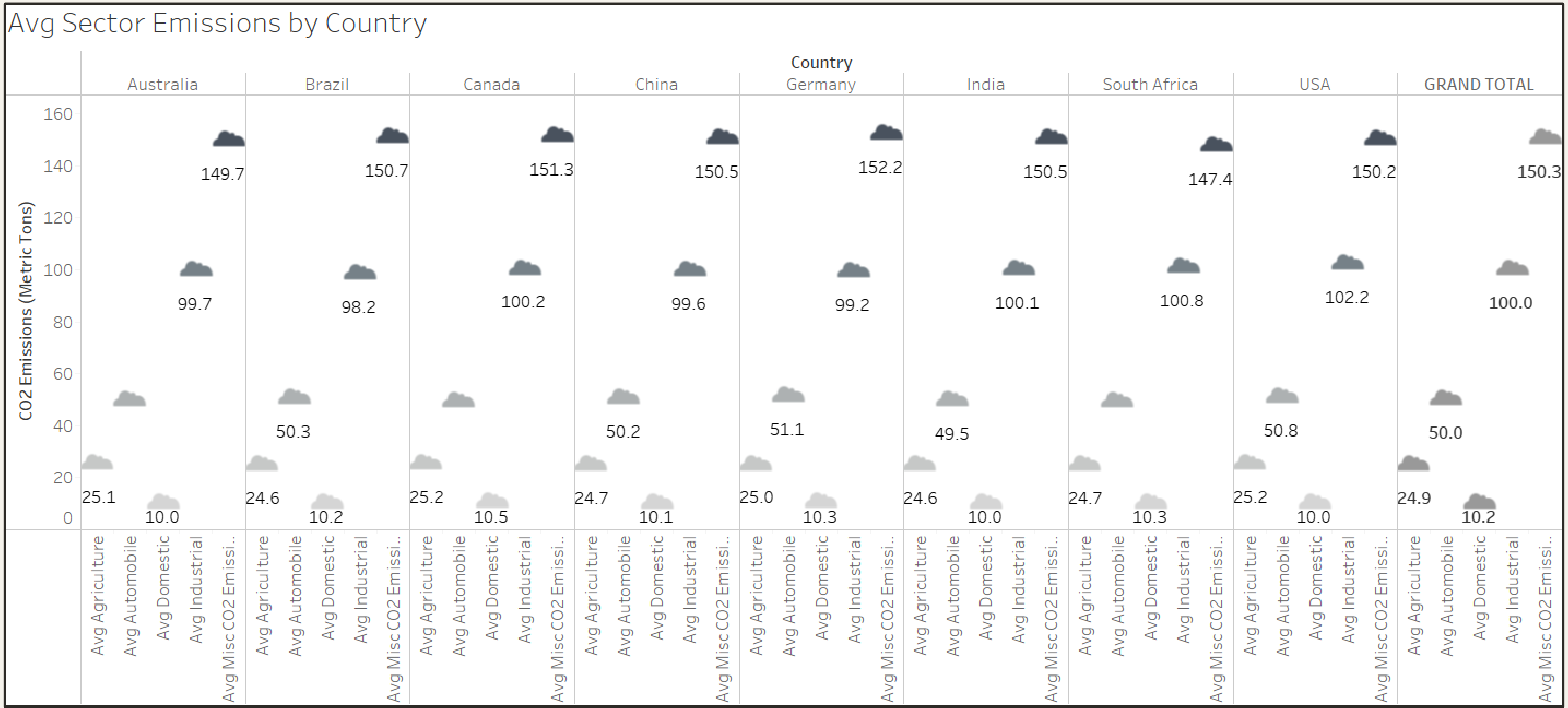
SECTOR CO2 EMISSIONS BY COUNTRY

The miscellaneous sector accounts for the most CO2 emissions and the domestic sector accounts for the least CO2 emissions in every single country.

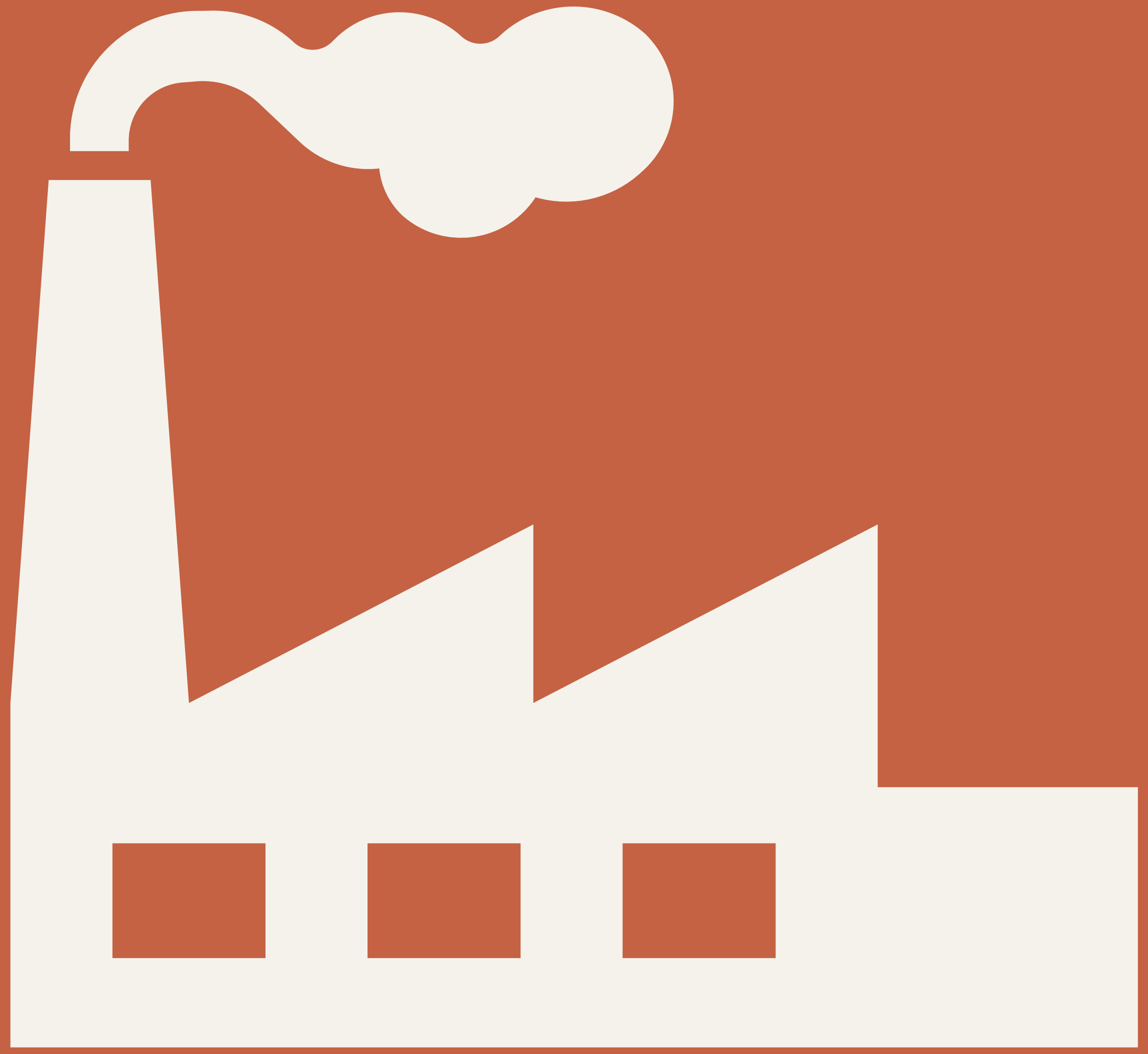


SECTOR CO2 EMISSIONS BY COUNTRY (AVERAGE)

On average every sector in every country emits relatively the same amount of CO2 emissions. As previously stated, the miscellaneous sector accounts for the most CO2 emissions on average in each country and the domestic sector accounts for the least CO2 emissions on average in each country.



CO2 EMISSIONS BY INDUSTRY



INDUSTRY CO2 EMISSIONS (AVERAGE)

On average every industry in every country emits relatively the same amount of CO2 emissions. The construction industry accounts for the most CO2 emissions at 337 metric tons on average and the mining industry accounts for the least CO2 emissions at 333 metrics tons on average.

Avg/Total Country - Industry Emissions

Industry Type	Country	Avg CO2 Emissions (Metric Tons)	Total CO2 Emissions (Metric Tons)
Construction	Australia	334	186,999
	Brazil	336	197,973
	Canada	335	189,057
	China	336	175,495
	Germany	339	178,643
	India	335	184,372
	South Africa	331	181,458
	USA	349	182,328
	Total	**337**	**1,476,325**
Energy	Australia	335	190,877
	Brazil	342	200,101
	Canada	342	173,980
	China	327	176,092
	Germany	343	195,924
	India	334	174,391
	South Africa	332	176,979
	USA	335	189,218
	Total	**336**	**1,477,564**
Manufacturing	Australia	333	185,092
	Brazil	333	187,513
	Canada	338	189,464
	China	339	191,676
	Germany	338	188,732
	India	338	195,969
	South Africa	327	177,877
	USA	336	197,693
	Total	**335**	**1,514,016**
Mining	Australia	334	187,192
	Brazil	326	196,605
	Canada	332	185,085
	China	339	181,011
	Germany	332	183,825
	India	332	180,526
	South Africa	340	185,124
	USA	334	163,827
	Total	**333**	**1,463,195**

INDUSTRY CO2 EMISSIONS (CONT.)

- Most Construction Emissions: Brazil at 197,973 metric tons
- Most Energy Emissions: Brazil at 200,101 metric tons
- Most Manufacturing Emissions: USA at 197,693 metric tons
- Most Mining Emissions: Brazil at 196,605 metric tons

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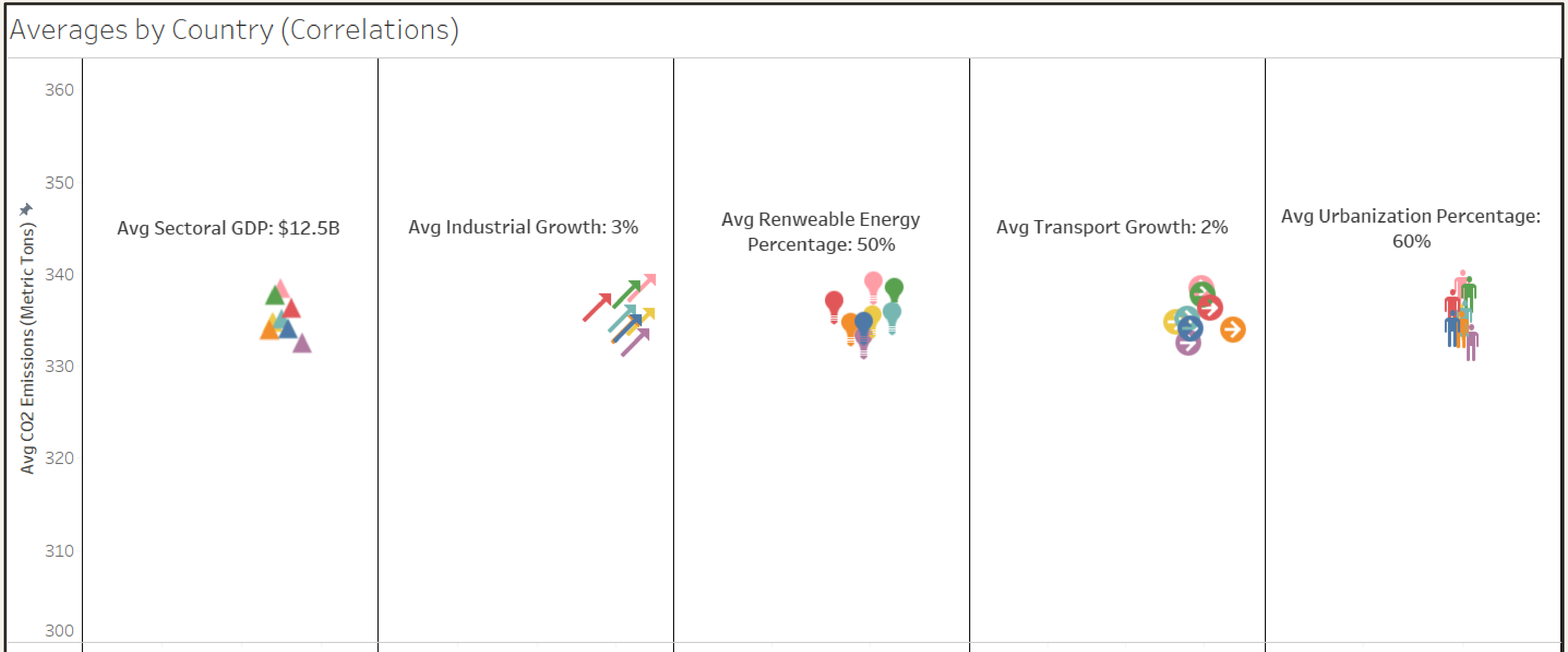
CO2 EMISSIONS CORRELATIONS

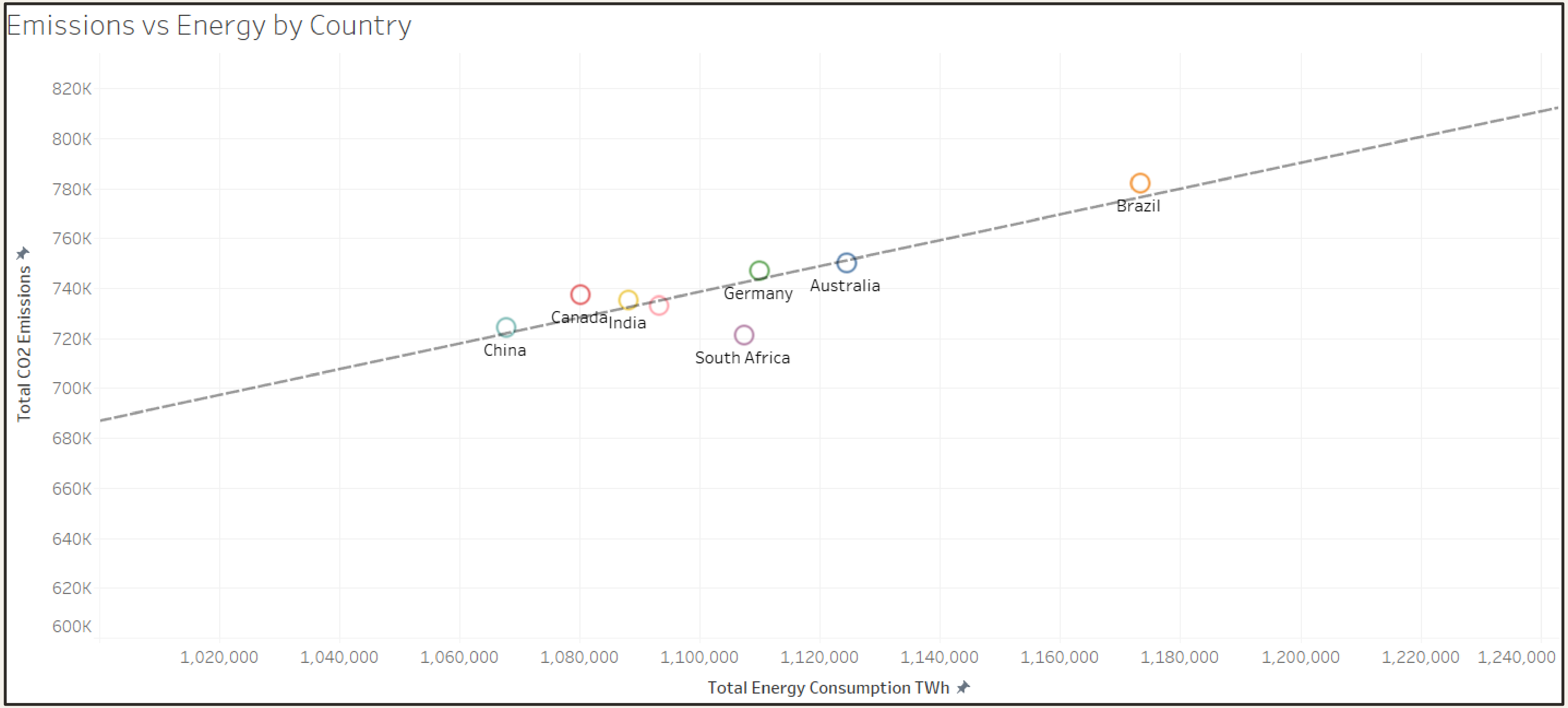
CO2 EMISSIONS AND ECONOMIC CORRELATION

Countries that produce a higher amount of CO_2 emissions (on average) tend to have a lower urbanization percentage, a lower industrial growth percentage, a high transport growth percentage, and a higher renewable energy percentage. Countries that produce less CO_2 emissions (on average) tend to have a higher sectorial GDP.



CO2 EMISSIONS AND ENERGY CORRELATIONS

Countries that have a high energy consumption rate tend to produce more CO2 emissions.



IN CONCLUSION...

FINDINGS

The Manufacturing (overall) and Construction (on average) industries contribute the most CO_2 emissions.

Miscellaneous CO_2 emissions is the sector that contributes the most CO_2 emissions (overall and on average).

The Mining industry contributes the least amount of CO_2 emissions (overall and on average).

Domestic CO_2 emissions have contributed the least amount of CO_2 emissions (overall and on average).

The average person emitted 1.8 metric tons of CO_2, contributing to a total of 5,913,101 metric tons of CO_2.

Brazil (overall) and USA (on average) seem to be the major contributors of CO_2 emissions.

South Africa seems to contribute the least amount of CO_2 emissions (overall and on average).

SUMMARY

- ✓ There does not seem to be a trend when it comes to CO_2 emissions over the years as the numbers tend to fluctuate in this regard.
- ✓ There is a downward trend when it comes to the amount of CO_2 emissions a country has and the average sectoral GDP
- ✓ There seems to be a slight negative correlation between the amount of CO_2 emissions a country produces and their average Industrial Growth Percentage.
- ✓ There is an upward trend when it comes to the amount of CO_2 emissions a country produces and a country's average Transport Growth Percentage.
- ✓ There is a negative correlation between the CO_2 emissions a country has produced and their average Urbanization Percentage.
- ✓ There is an upward trend when it comes to CO_2 emissions a country produces and a country's average Renewable Energy Percentage.
- ✓ Overall, we can assume that more CO_2 emissions are a result of more Energy Consumption.

FURTHER ANALYSIS

- I would like to get more data from previous years to do an overall assessment of CO_2 emissions.
- I would like to have every country's CO_2 emissions information.
- More data on deforestation to see the influence of CO_2 emissions.
- I would like to have some natural disaster data to compare CO_2 emissions.
- I would like to get detailed information about what other sectors make up the Miscellaneous CO_2 emissions attribute. I feel like we can get even more granularity as to what sectors are causing high levels of CO_2 emissions.
- I would also like to have some type of climate temperature measurements for each country to compare the CO_2 emissions.